Cliques with Common Pendant Leaves: A Positive Family for the Chromatic-Polynomial Graphon Bound

Abstract

Let $H_{r,k}$ be obtained from K_r by attaching k leaves to one fixed clique vertex. For every $r \geq 3$, $k \geq 0$, and every graphon W of edge density

$$p \geq 1 - \frac{1}{r-1},$$

we prove

$$t(H_{r,k}, W) \geq p^k \prod_{j=0}^{r-1} \big(1 - j(1-p)\big) = \Phi_{H_{r,k}}(p).$$

This is the complete density interval required by $\chi(H_{r,k}) = r$. The proof is a direct link-graphon argument. It does not assume a general universal-vertex or join closure. In the catalogue on at most six vertices it proves the previously open Atlas graphs 45, 133, and 191.

1 The family and its exact target

Let (Ω, μ) be a probability space and let $W\colon \Omega^2 \to [0,1]$ be a symmetric measurable graphon. For a finite simple graph F, write

$$t(F, W) = \int_{\Omega^{V(F)}} \prod_{uv \in E(F)} W(x_u, x_v) \prod_{u \in V(F)} \mathrm{d}\mu(x_u).$$

This is the ordinary non-injective homomorphism density. Put

$$p = t(K_2, W), \qquad d(x) = \int_\Omega W(x,y)\,\mathrm{d}\mu(y).$$

For integers $r \geq 3$ and $k \geq 0$, define $H_{r,k}$ as follows. Start with a clique K_r, distinguish one of its vertices as o, add new vertices $u_1, \dots, u_k$, and add precisely the new edges ou_i. Thus all k new vertices are leaves with the same neighbour o.

After a proper colouring of K_r has been chosen, each leaf has $x-1$ choices. Therefore

$$\chi_{H_{r,k}}(x) = (x)_r (x-1)^k, \qquad \chi(H_{r,k}) = r, \tag{1}$$

where $(x)_r = x(x-1)\cdots(x-r+1)$. Define

$$A_r(p) := \prod_{j=0}^{r-1} \big(1 - j(1-p)\big).$$

If $q = 1 - p$, then

$$\begin{aligned}\Phi_{H_{r,k}}(p) &= q^{r+k} \left(\frac{1}{q}\right)_r \left(\frac{p}{q}\right)^k \\ &= p^k A_r(p). \end{aligned} \tag{2}$$

The identity extends continuously to $p = 1$.

2 The accepted clique input and its tangent lines

We use one previously accepted result: the clique case of the pure-chordal theorem. In the notation above it says the following.

Theorem 2.1 (Clique-density input)**.** *Let $m \geq 2$. Every graphon V of edge density z satisfying*

$$z \geq a_m := 1 - \frac{1}{m-1}$$

satisfies

$$t(K_m, V) \geq A_m(z).$$

For $m = 2$, the convention $a_2 = 0$ is used, and the assertion is the identity $t(K_2, V) = z = A_2(z)$.

For completeness, the input for $m \geq 3$ is exactly the specialization of the pure-chordal formula to the single maximal clique K_m. It can also be obtained by multiplying the Moon–Moser clique-ratio inequalities

$$\frac{t(K_j, V)}{t(K_{j-1}, V)} \geq 1 - (j-1)(1-z), \qquad 2 \leq j \leq m,$$

whose right-hand sides are nonnegative when $z \geq a_m$.

Lemma 2.2 (Clique-target shape)**.** *For $m \geq 2$, the polynomial*

$$\phi_m(z) := A_m(z)$$

is nonnegative, nondecreasing, and convex on $[a_m, 1]$, and $\phi_m(a_m) = 0$.

Proof. Ignoring the constant factor corresponding to $j = 0$, write

$$\phi_m(z) = \prod_{j=1}^{m-1} \ell_j(z), \qquad \ell_j(z) = 1 - j(1-z).$$

Every ℓ_j is affine and nondecreasing, and every ℓ_j is nonnegative on $[a_m, 1]$. The product is therefore nonnegative. Its first derivative is a sum of products of nonnegative factors, and its second derivative is a sum over ordered pairs of distinct differentiated factors, again with only nonnegative factors. Hence $\phi_m' \geq 0$ and $\phi_m'' \geq 0$ on this interval. At a_m, the last factor ℓ_{m-1} is zero. □

Lemma 2.3 (Global tangent bound)**.** *Let $m \geq 2$ and fix $c \in [a_m, 1]$. Every graphon V, of arbitrary edge density $z \in [0, 1]$, satisfies*

$$t(K_m, V) \geq \phi_m(c) + \phi_m'(c)(z - c).$$

Proof. If $z \geq a_m$, Theorem 2.1 and convexity give

$$t(K_m, V) \geq \phi_m(z) \geq \phi_m(c) + \phi_m'(c)(z - c).$$

If $z < a_m$, convexity and $\phi_m(a_m) = 0$ show that the tangent line at c is at most zero at a_m. Its slope is nonnegative, so it is also at most zero at $z < a_m$. The desired conclusion then follows from $t(K_m, V) \geq 0$. □

3 Weighted link density

Define the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Lemma 3.1 (Weighted rooted-triangle inequality)**.** *Suppose $p > 0$. For every integer $h \geq 0$,*

$$\int_\Omega d(x)^h \tau(x)\,\mathrm{d}\mu(x) \geq \frac{2p-1}{p} \int_\Omega d(x)^{h+2}\,\mathrm{d}\mu(x).$$

Proof. Put $M_j = \int d^j$, let T_W be the graphon operator, and write

$$I_h = \int d^h \tau, \qquad J_h = \left\langle d^h, T_W d \right\rangle.$$

The elementary inequality $ab \geq a + b - 1$ for $a, b \in [0,1]$, applied to $W(x,y)$ and $W(x,z)$ and then multiplied by $d(x)^h W(y,z)$, gives

$$I_h \geq 2J_h - pM_h. \tag{3}$$

Set $U = 1 - W$, $u = T_U \mathbf{1} = 1 - d$, and $q = 1 - p$. Since

$$T_W d = p\mathbf{1} - T_U d = d - q\mathbf{1} + T_U u,$$

we have

$$J_h = M_{h+1} - qM_h + \left\langle d^h, T_U u \right\rangle. \tag{4}$$

Symmetry of U gives

$$\begin{aligned} &2\left\langle d^h, T_U u \right\rangle - 2\int_\Omega d(x)^h u(x)^2\,\mathrm{d}\mu(x) \\ &\quad = \int_{\Omega^2} U(x,y)\big(u(y) - u(x)\big)\big(d(x)^h - d(y)^h\big)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \geq 0. \end{aligned}$$

Indeed, $d = 1 - u$, so u and d^h are oppositely ordered. Substitution in (4), using $u = 1 - d$, yields

$$J_h \geq M_{h+2} - M_{h+1} + pM_h.$$

Together with (3), this gives

$$I_h \geq 2M_{h+2} - 2M_{h+1} + pM_h.$$

Finally,

$$\begin{aligned} &2M_{h+2} - 2M_{h+1} + pM_h - \frac{2p-1}{p} M_{h+2} \\ &\quad = \frac{1}{p}\int_\Omega d(x)^h (d(x) - p)^2\,\mathrm{d}\mu(x) \geq 0. \end{aligned}$$

□

For $d(x) > 0$, define the probability measure

$$\mathrm{d}\mu_x(y) = \frac{W(x,y)}{d(x)}\,\mathrm{d}\mu(y)$$

and let W_x denote the same kernel W, now viewed on (Ω, μ_x). Its edge density is

$$z_x = t(K_2, W_x) = \frac{\tau(x)}{d(x)^2}.$$

At points where $d(x) = 0$, put $z_x = 0$.

Corollary 3.2 (Weighted average link density)**.** *For every integer $n \geq 2$ and $p > 0$,*

$$\int_\Omega d(x)^n z_x\,\mathrm{d}\mu(x) \geq \left(2 - \frac{1}{p}\right)\int_\Omega d(x)^n\,\mathrm{d}\mu(x).$$

Proof. The left-hand side is $\int d^{n-2}\tau$. Apply Lemma 3.1 with $h = n - 2$. □

4 Main theorem

Theorem 4.1 (Cliques with common pendant leaves)**.** *Let $r \geq 3$ and $k \geq 0$. Every graphon W with edge density*

$$p \geq 1 - \frac{1}{r-1}$$

satisfies

$$\boxed{t(H_{r,k}, W) \geq p^k A_r(p) = \Phi_{H_{r,k}}(p).}$$

This is the full interval required by $\chi(H_{r,k}) = r$.

Proof. Put

$$m = r - 1, \qquad n = r - 1 + k, \qquad c = 2 - \frac{1}{p}.$$

The assumed lower bound on p implies

$$c \geq 1 - \frac{1}{r-2} = a_m.$$

Also $c \leq 1$. Let $M_n = \int d^n$.

Condition on the image x of the distinguished clique vertex o. The other $r - 1$ clique vertices form K_m inside the link graphon W_x, while the k leaves become isolated vertices there. Directly from the homomorphism integral,

$$t(H_{r,k}, W) = \int_\Omega d(x)^n t(K_m, W_x)\,\mathrm{d}\mu(x). \tag{5}$$

No division is needed on the set $d(x) = 0$, because $n \geq 2$ and its contribution to (5) is zero.

Apply Lemma 2.3 to each W_x, with tangent point c:

$$t(K_m, W_x) \geq \phi_m(c) + \phi_m'(c)(z_x - c).$$

Multiplying by $d(x)^n$ and integrating gives

$$t(H_{r,k}, W) \geq M_n \phi_m(c) + \phi_m'(c) \left(\int d(x)^n z_x \, \mathrm{d}\mu(x) - cM_n \right).$$

The second line is nonnegative by Corollary 3.2 and Lemma 2.2. Hence

$$t(H_{r,k}, W) \geq M_n \phi_m(c).$$

Since $\phi_m(c) \geq 0$, Jensen's inequality gives

$$t(H_{r,k}, W) \geq p^n \phi_m(c).$$

It remains only to identify this expression. If $q = 1 - p$, then $1 - c = q/p$, so

$$\begin{aligned} \phi_m(c) &= \prod_{j=0}^{r-2} \left(1 - \frac{jq}{p}\right) \\ &= p^{-(r-1)} \prod_{j=0}^{r-2} (p - jq) \\ &= p^{-(r-1)} \prod_{\ell=1}^{r-1} (1 - \ell q) = p^{-(r-1)} A_r(p). \end{aligned}$$

As $n = r - 1 + k$, it follows that

$$p^n \phi_m(c) = p^k A_r(p) = \Phi_{H_{r,k}}(p),$$

where the last equality is (2). □

5 Thresholds and equality

At the required threshold

$$p = 1 - \frac{1}{r-1},$$

the last factor of $A_r(p)$ is zero. Thus the target is zero, and the inequality also follows directly from nonnegativity. The balanced complete $(r-1)$-partite graphon attains equality there because it has no copy of K_r.

More generally, for every integer $\ell \geq r - 1$, the balanced complete ℓ-partite graphon T_ℓ has edge density $p = 1 - 1/\ell$ and

$$t(H_{r,k}, T_\ell) = \frac{\chi_{H_{r,k}}(\ell)}{\ell^{r+k}} = \Phi_{H_{r,k}}(p).$$

At $p = 1$, one has $W = 1$ almost everywhere, and both sides equal 1.

6 Catalogue consequences

Corollary 6.1 (New positive Atlas cases)**.** *The following previously open graphs satisfy the desired inequality on their full required density intervals:*

Atlas ID	*graph6*	*family member*	*target*
45	`DJ{`	$H_{4,1}$	$pA_4(p)$
133	`E~a?`	$H_{4,2}$	$p^2A_4(p)$
191	`E~|?`	$H_{5,1}$	$pA_5(p)$

Thus Atlas 45 and 133 are positive for $p \geq 2/3$, while Atlas 191 is positive for $p \geq 3/4$.

Proof. Atlas 45 is K_4 with one leaf, Atlas 133 is K_4 with two leaves having the same clique neighbour, and Atlas 191 is K_5 with one leaf. The graph6 codes are independent machine-readable identifiers. Apply Theorem 4.1. □

7 Scope of the proof

The only previously accepted graph-density result used here is the clique case of the pure-chordal theorem, stated precisely as Theorem 2.1. The weighted rooted-triangle estimate, the tangent extension, and the link averaging are all proved in this note.

Although the proof conditions on the distinguished clique vertex, it does not assert that adjoining a universal vertex preserves positivity for an arbitrary graph. It uses the explicit clique polynomial A_{r-1}, its verified convexity, and the exact clique-density input. Thus it establishes only the stated common-leaf clique family, not a general join or lifting closure.

All calculations apply directly to arbitrary graphons. They require no regularity, finite-step reduction, injective-copy interpretation, minimum-degree bound, or pointwise lower bound on $d(x)$.

A What the Lean formalization actually proves

Theorem 4.1 is formalized as stated, for all $r \geq 3$ and $k \geq 0$; Atlas 45, 133 and 191 are `verified`. The development is `lean/Taeyoung/Methods/CliqueLeaf/`. The proof follows the note step for step — Theorem 2.1, Lemma 2.3, Lemma 3.1, Corollary 3.2 and the link identity (5) are each a named theorem there. Two remarks.

Theorem 2.1 is proved, not accepted

The clique input is the one previously accepted result the note uses. It is formalized, as the clique case of the pure-chordal development (`Methods/PureChordal/CliquePolynomialBound.lean`), so no accepted graph-density result enters the catalogue through this family.

The chain is stated once, abstractly

Nothing in the argument uses the clique polynomial beyond four facts: it vanishes at the threshold, it has nonnegative slope, its tangent at c lies under it on $[a, 1]$, and it bounds the base density above the threshold. The Lean development therefore states the chain once with the polynomial and the base graph abstract (`Methods/ConeBound.lean`), and this family is its first instance. Three later families — the paw and triangle–edge cones, the verified-base cones, and the clique join of C_5 — are further instances of the same theorem, so the argument below is not repeated anywhere.